

Exploring Bias in the New York Times Article and its Effects on Society

Final Report

Introduction

In a time close to a breaking point of our world, we look to the people we trust for information. Our world is currently filled with a vast amount of injustice, wars, and darkness, making it difficult to keep up with everything. Since the balance of good and evil is so delicate, we trust those in our community to make those distinctions for us. Our community nowadays come in the form of news sources that summarize everything we need to know about a specific event in our world.

News sources were created to collect and distribute information to their audience without expressing a preference or bias when conveying this information. The idea behind these news sources is to provide their audience with facts and details, allowing them to form their own opinions and thoughts based on the information provided. However, the issue arises when the news source incorporates biases into the delivery of these facts and details, as the goal of these sources then becomes different.

The biggest of these news source outlets is the New York Times, an American daily newspaper based in New York City. One of the New York Times values is independence, where they state that they have pledged “to give the news impartially, without fear or favor, regardless of party, sect, or interests involved.”¹ However, many believe that the Times has gone against its pledge, believing they allude to their reader’s emotions to affect their perspective in a given issue.

A great example of this bias falls when reporting about the Palestine-Israel dispute, where many believe that the Times reporting falls more into Israel’s side. Making Israel appear as a good, just, and pure system while Palestine is an evil, unjust, and corrupt system. In order to explore the biases of this topic, it is important to understand it.

The Palestine-Israel situation started a long time ago, but one of the most important highlights was in 1948. In 1948, Israel was established, however, the establishment of Israel was built on an already populated land that was called Palestine, which was under the British mandate before the creation of Israel, and was part of the Ottoman Empire before that. The dispute over the land lasts till today, and there are many instances where news sources were proven to be biased to Israel’s side. For the relevancy of biases this paper wanted to explore, it was decided to explore recent events that have occurred in the Palestine-Israel issue. On October 7th, Israel was breached from a militant group called Hamas, and approximately 1200 Israelis were killed in that attack. Israel’s response was a ground invasion of the Palestinian city of Gaza, where the militant group resides. Due to this extreme response and the indifference killing between a militant and a civilian, many labelled this response as a Genocide, and it is estimated that about 45,000 Palestinians have been killed since October 7th².

Taking up all the facts on this issue, this paper **explores biases in the New York Times Articles, as it is vital for news to be impartial.**

Objective

This projects aims to analyze the factors affecting biases analyzing the article's headlines and some of their bodies. Exploring many trends in these headlines through sentiment analysis, polarity analysis, and biases analysis.

Key Findings

- Approximately 86% of New York Times article are pro-Israeli, while 14% are pro-Palestinian.
- Headlines with high sentiment scores consistently demonstrated high bias scores, showing a strong correlation between emotional language and partisan coverage.
- Regarding Israel, headlines related to ceasefire topics had the highest negative sentiment scores.
- Israel headlines with the highest positive sentiment scores occurred when discussing the country's responses to enemies and military achievements.
- Regarding Palestine-related headlines, they displayed their highest negative sentiment scores when discussing the Palestinian government or using graphic vocabulary.
- Palestine-related headlines had their highest positive sentiment scores when discussing truces and reconstruction efforts.

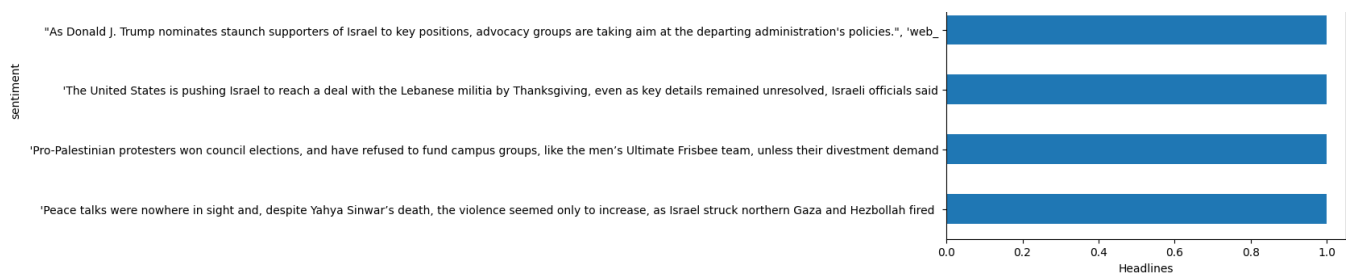
Methodology Overview

The data was collected using the New York Times API³, specifically three keywords were used in the query (Palestine, Israel, and Gaza) to observe the differences in analysis between the headlines that contained these keywords. In order to explore biases, Dbias (python library) was utilized. Additionally, TextBlob (python library) was utilized to explore sentiments and polarity. Leveraged charts such as line charts and bar charts to show sentiment scores, bias probability, and the relationship between Israeli and Palestinian casualties. Through the combination of using newsAPI⁴, Aljazeera News^{aljazeeraBreakirengNews?}, and The Jerusalem Post⁵. Lastly used models such as Naive Bayes model, Random Forest Classifier, and Decision Tree Classifier for supervised learning while utilized a K-means model for unsupervised training.

Visualisations

³Israeli attacks hit air-defense systems around important energy sites but avoided the facilities themselves, Iranian and Israeli officials said. The United St



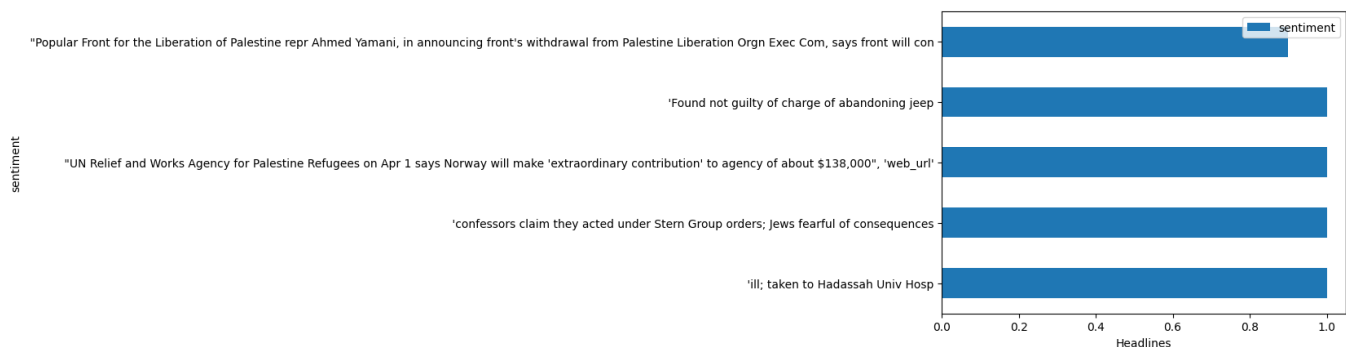


Plot 1: Top 5 Highest Sentiment Score When Israel was Mentioned

The following plot shows the top 5 headers that had the highest sentiment score when Israel was mentioned. Some headers were cut in order for the plot to have a better visualisation. The first header that is seen is “Peace talks were nowhere in sight and, despite Yahya Sinwar’s death, the violence seemed only to increase, as Israel struck northern Gaza and Hezbollah fired dozens of projectiles.” The biases can be seen when the Times talks about what Israel is doing, with words such as “struck” in comparison to the militant group Hezzollah who “fired dozens of projectiles.” This bias is created to allude to the people’s emotions, where there would be no anger or strong feelings towards Israel, as struck can mean a lot of things, in contrast to the feelings towards Hezbollah, where there are more details to what they have done making the reader feel anger.

The second highest is “Pro-Palestinian protesters won council elections, and have refused to fund campus groups, like the men’s Ultimate Frisbee team, unless their divestment demand.” The Times uses a classical technique of putting attention on Pro-Palestinians by having the term be the first thing the audience sees, and make comparisons of campus groups, like the men’s Ultimate Frisbee team, a term that has nothing to do with the issue to signify what this group is doing is affecting others, this comparison is aimed to make the audience stand against this group and their divestment demand since they are not funding innocent groups, taking away from the actual reason that this group is protesting.

The third highest, “The United States is pushing Israel to reach a deal with the Lebanese militia by Thanksgiving, even as key details remain unresolved.” The Times conveys a fact about what is currently going on with Israel, however, it throws shade at the United States by undermining the deal it is trying to make with the Lebanese militia. This is done by including the last part of the header, explaining that this deal does not make sense as there are key details that are still unresolved.



Plot 2: Top 5 Highest Sentiment Score When Palestine was Mentioned

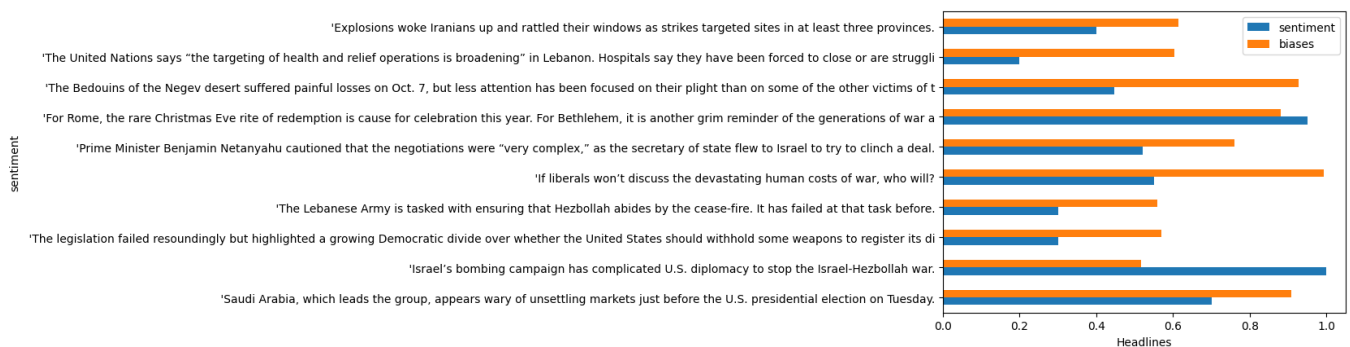
The first header relates to the article “Gillette in Palestine Hospital.” The reason why it is classified as biases here is not as easy to detect as when Israel is mentioned. This may be because the sentiment here is positive rather than negative as before, meaning the bias is positive as it is telling the audience

about something positive. Unfortunately this article is not accessible, as it was written by Times in 1946, which could be the reason for a good bias in a topic concerning Palestine, as Israel has been established for a year then.

Second header is relating to an article named “Palestine Jews Fearful”, from here Times can be seen to allude to the people’s emotions in sympathizing with Palestinian Jews. This was a shocking discovery, as it is rare to see the Times alluding to people’s emotions to sympathize with Palestinians. After more in depth analysis of the article, it was discovered to be written in 1944, before the establishment of Israel, which gives a pretty good reason for why the Times was biased towards Palestinians then.

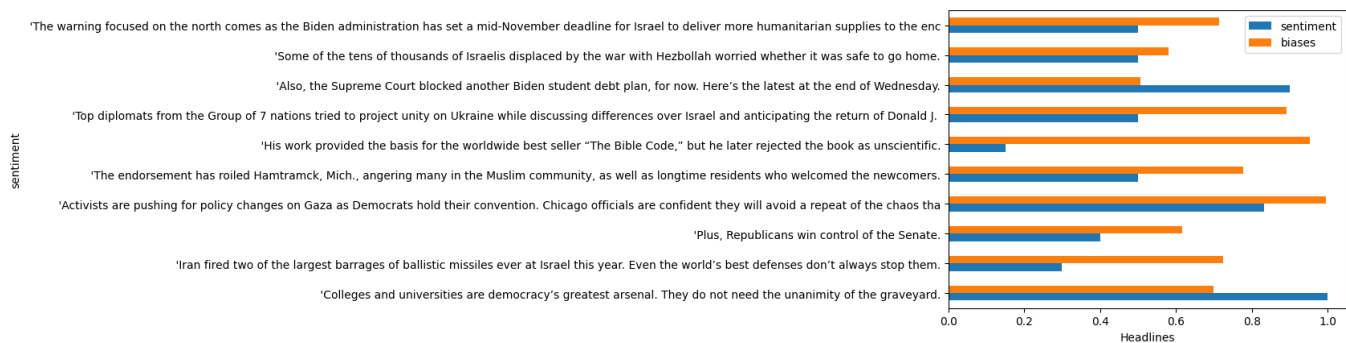
The third header is “UN Relief and Works Agency for Palestine Refugees on Apr 1 says Norway will make ‘extraordinary contribution’ to agency of about \$138,000.” The biases here is harder to detect than when Israel is mentioned, showing how noticeable it is when either Palestine or Israel is mentioned. The biases here could be from the words “extraordinary contribution” which are detected as biased, but as we can see from the headline this biases does not come from the Times but rather from a quote extracted from the “UN Relief and Works Agency.”

All and all, we can see a very noticeable distinction between the sentimental analysis of the headers and snippets of the Times articles when Israel and Palestine are mentioned. As when the sentiment score was high when Israel was mentioned, it was easier to detect biases than when the sentiment score was high when Palestine was mentioned. The last observation made is noticing that when the sentiment value and the biases were high when Palestine was mentioned, they were for headers that were around the Time Israel was still establishing itself as a country.



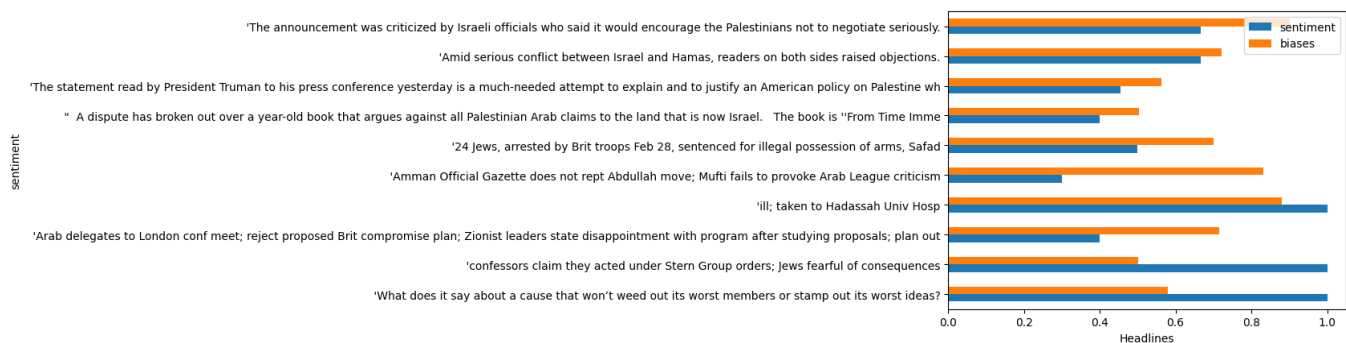
Plot 3: Top 10 Lowest Polarity Score When Israel was Mentioned.

Polarity in this context relates to whether a sentiment value is positive or negative about a topic. From the plot above, we can see that when the author’s sentiment was in a negative tone when Israel was mentioned, it was only when talking about a ceasefire or peace, making it seem like peace was a horrible solution to the issue concerning Israel. For example, the header, “The Lebanese Army is tasked with ensuring that Hezbollah abides by the cease-fire. It has failed at that task before”, conveys a terrible impression to the audience about a ceasefire with Hezbollah, since the Lebanese army failed to make Hezbollah abide by the ceasefire, then why should Israel try again? Such ideas that the Times conveys to its readers insight very strong emotions that pushes the readers to not support a path towards peace. A classic tactic that some news sources may use to keep the audience engaged with their articles.



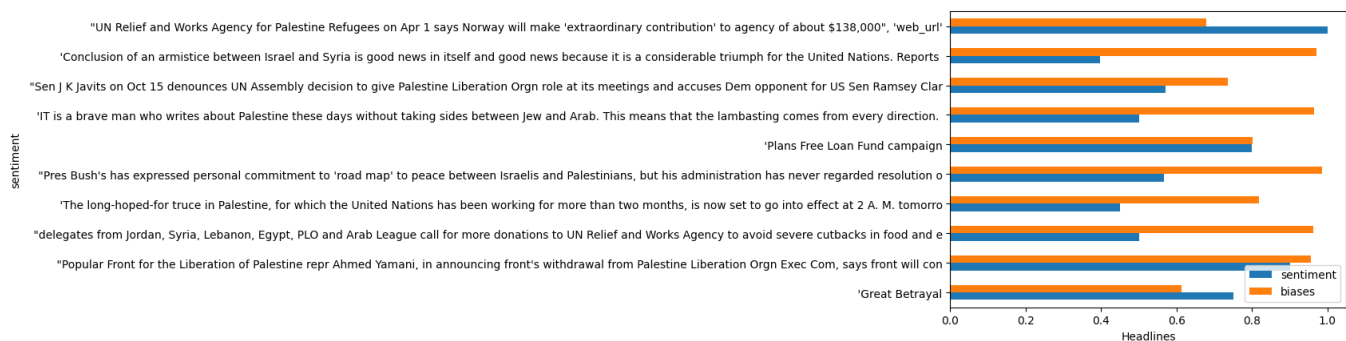
Plot 4: Top 10 Highest Polarity Score When Israel was Mentioned.

From the plot above, it can be seen that when the author's sentiment was in a positive tone when Israel was mentioned, it was when talking about Israel's enemies, more specifically with how Israel deals with its enemies. For example, the header, "Iran fired two of the largest barrages of ballistic missiles ever at Israel this year. Even the world's best defenses don't always stop them", the times clearly pushes its audience to have strong emotions towards Iran, where it uses words such as largest, barrages, and ballistic to convey fear and shock towards their audience. The Times tries to undermine Iran's attack by promoting Israel's defenses, by using words such as "best", conveying that Israel did good to defend itself against such a "intense" attack.



Plot 5: Top 10 Lowest Polarity Score When Palestine was Mentioned.

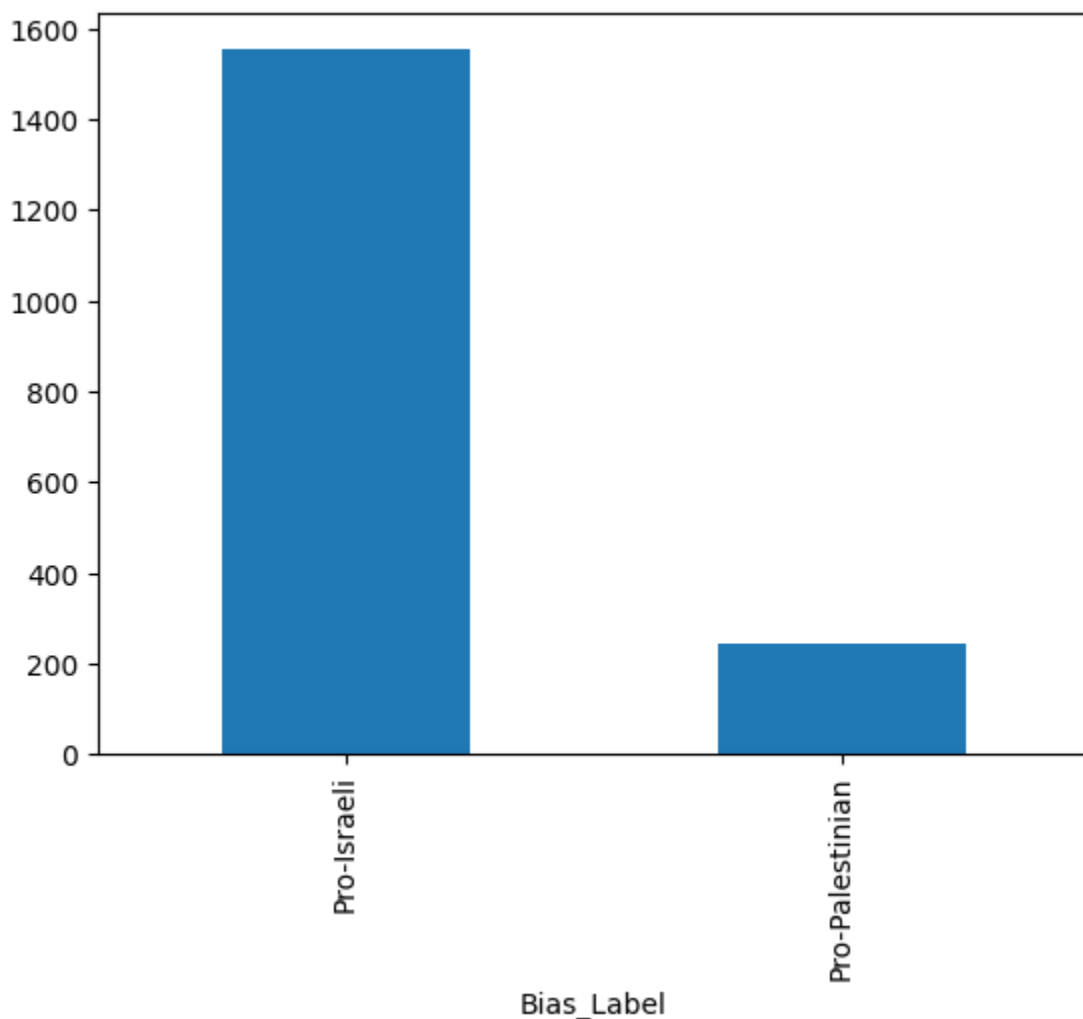
From plot 5 it can be that when the polarity is negative, then the author is criticizing Palestine. For example, the top header is "What does it say about a cause that won't weed out its worst members or stamp out its worst ideas?" The Times criticizes the Palestinian government because of a few seeds, however, that makes sense for a sentiment when the polarity is negative, and is expected to be consistent when the Times discusses any topic. On the other hand, from the examples discussed before, it can be seen that is not the case.



Plot 6: Top 10 Highest Polarity Score When Palestine was Mentioned.

From plot 6, when the polarity is positive, the most common topics are about peace, a ceasefire, and rebuilding. For example, the header “The long-hoped-for truce in Palestine, for which the United Nations has been working for more than two months, is now set to go into effect at 2 A. M. tomorrow.” When the polarity is positive the subject is positive and under a good atmosphere that does not push the audience into picking a side, but rather be happy about a catastrophic topic and give the audience hope.

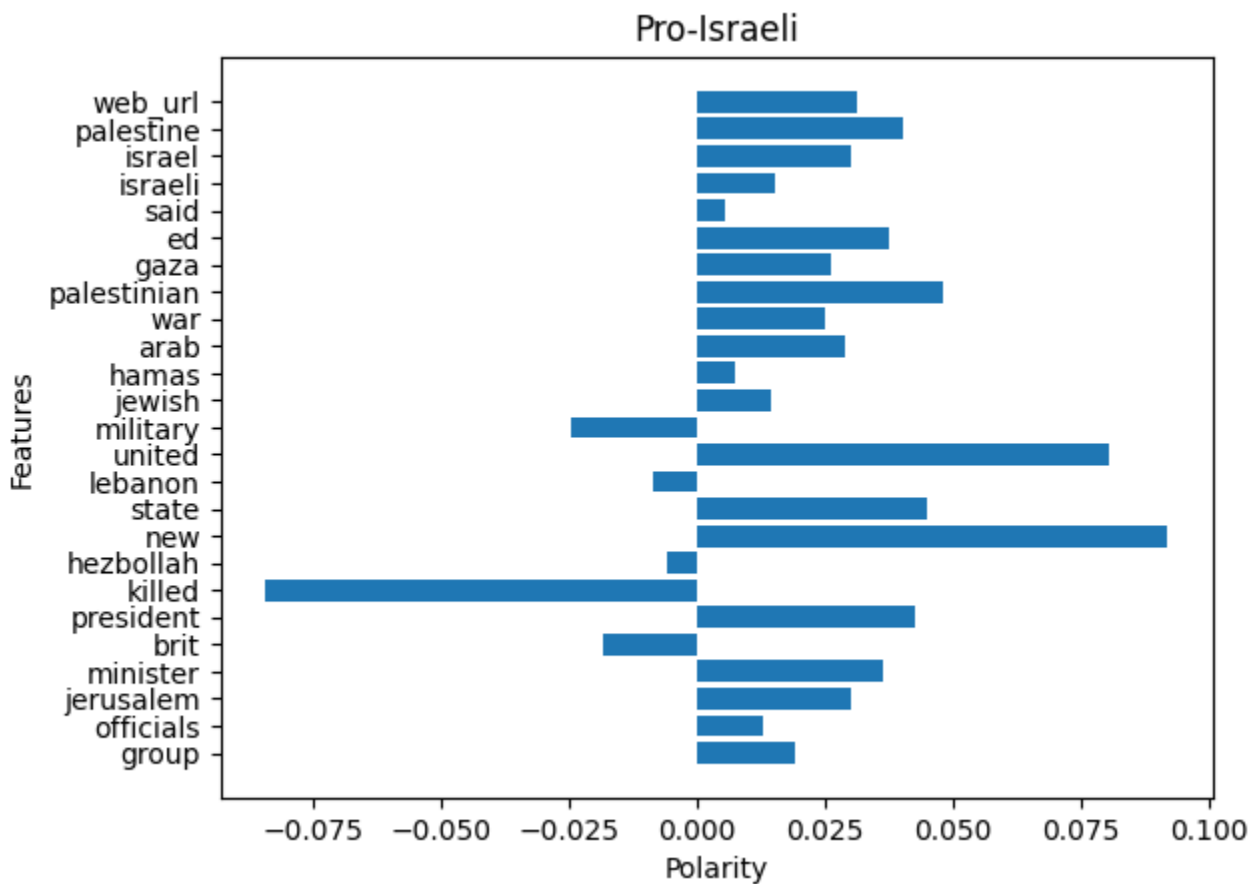
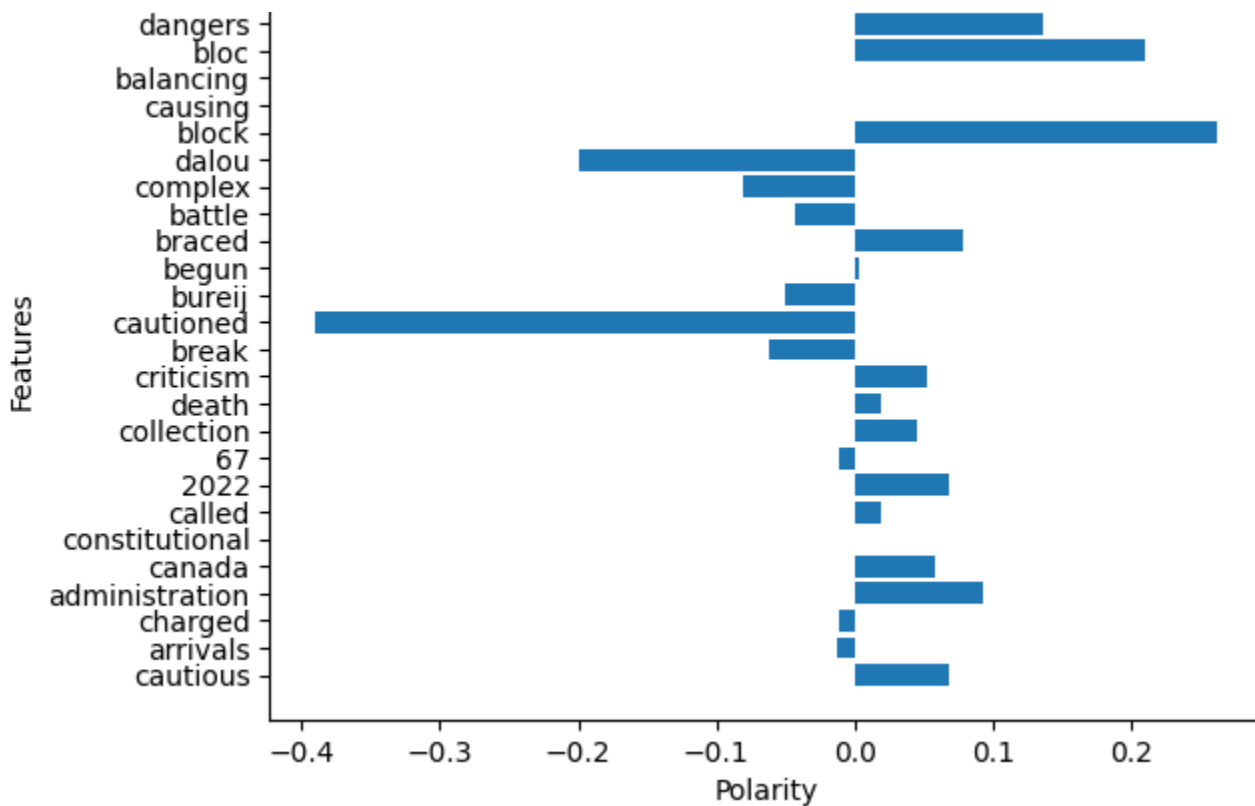
These plots make up a lot of interesting observations. When the polarity is negative the author attacks the solution of a ceasefire rather than Israel, although when it comes to Palestine the author attacks their government. Moreover, when the polarity is positive the author promotes Israel’s progression towards the war, but when it comes to Palestine they talk about a ceasefire and peace. One can conclude that the Times is very hypocritical and inconsistent when it comes to discussing either side, but it was decided to further analyse this.



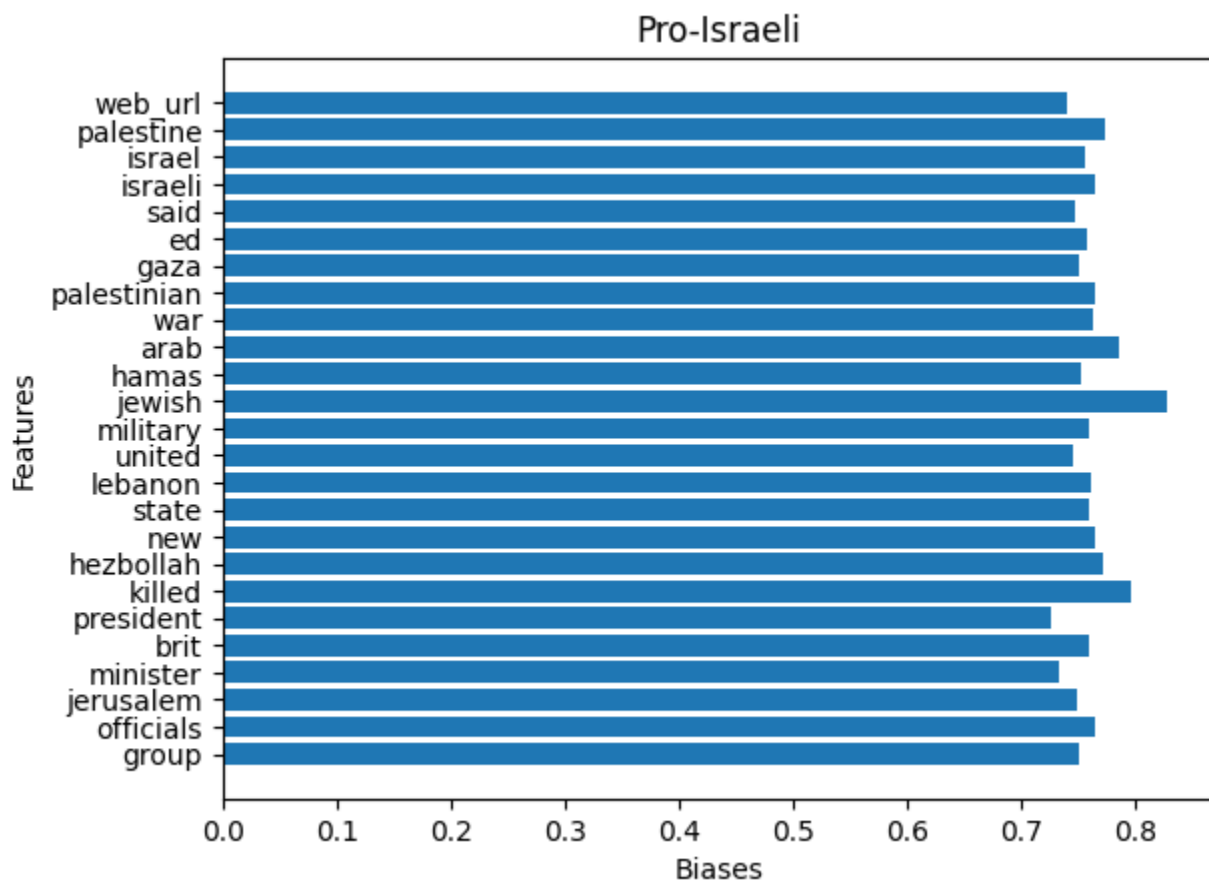
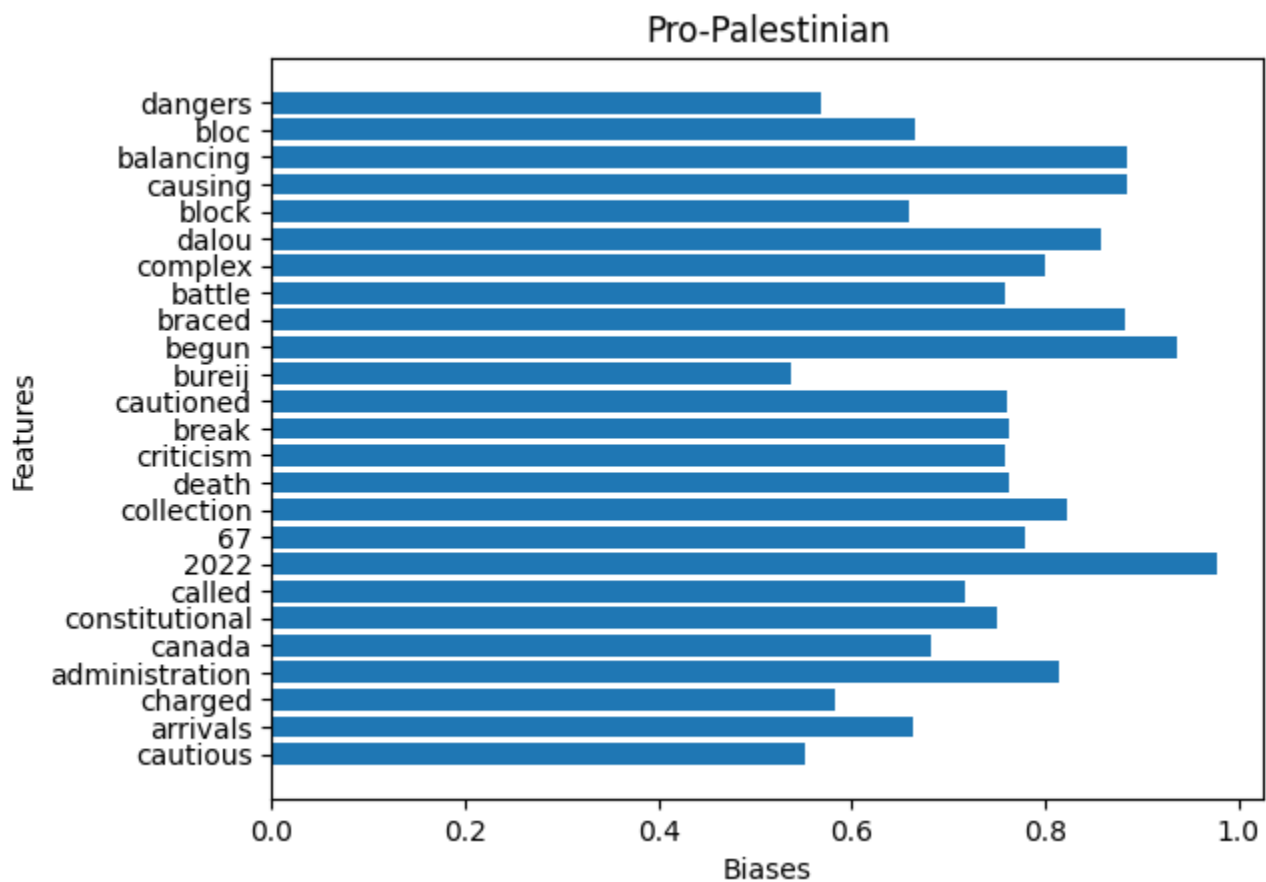
Plot 7: Count Rate of Pro-Israeli vs Pro-Palestinian labels in New York Times Headers.

About 86% of New York Times content about this issue leans toward Pro-Israeli, while 14% of the content leans toward Pro-Palestinian.





The words when the headers are labelled as Pro-Palestinian are verbs and have almost no polarity. On the other hand, when the headers are labelled as Pro-Israeli there are more sentimental values and nouns are more mentioned in the headers.



The difference of biases in the two categories can be seen, for words associated with Pro-Israeli headers the bias is always high and above 0.7, while for Pro-Palestinian categories it differs.

Societal Implications

This study clearly highlights the effect of biases in news media. These sources use many different tactics to keep the audience engaged, as they play on the audience's emotions using many triggering words.

Call to Action

The recommendation calls for delivering news in a consistent unbiased manner, where the audience can create their observations and opinions on their own without being influenced by the source itself.

Conclusion

Overall, it is critical for news to be impartial; however there is clear evidence of **biases in the New York Times Articles**. After analyzing the data, it was observed that when Israel was mentioned, the headlines of the NYT had a higher sentimental, polarity, and bias score than when Palestine was mentioned. Additionally, it can be seen that topics related to Israel with a negative tone were related to ceasefires. However, when Palestine was mentioned with a negative tone, the government was criticized. When the tone was positive, Israel was praised for its war outcomes, while for Palestine, ceasefire and peace were mentioned.

After using machine learning to determine whether a headline was leaning more towards being pro-Israeli or pro-Palestinian, it was concluded that about 86% of the headlines leaned towards being pro-Israeli. It was observed from using unsupervised learning that the top features related with Palestine were verbs that had varying bias scores and low polarity score. On the other hand, when Israel was mentioned, the top features were nouns with high bias and polarity scores. This implies that the NYT influences its readers to associate specific nouns with either positive or negative news related to Israel.

In conclusion, the analysis in this paper demonstrates that the New York Times, to some degree, manipulates its readers by associating certain nouns with negative or positive news about Israel. This issue can have a detrimental effect on people seeking to educate themselves on current events, as the biases present in certain headlines may influence an individual's perspective on a particular subject. Ultimately, this paper emphasizes the importance of having an unbiased title, allowing the reader to form their own opinions about a given topic without undue influence.

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